

Ear Discharge

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Ear discharge (otorrhea) is drainage from the ear. The drainage may be watery, bloody, or thick and whitish, like pus (purulent). Depending on the cause of the discharge, people may also have [ear pain](#), fever, itching, [vertigo](#), ringing in the ear ([tinnitus](#)), and/or [hearing loss](#). Symptoms range from sudden and severe to slowly developing and mild.

Causes of Ear Discharge

Discharge may originate from the ear canal, the middle ear, or, rarely, from inside the skull.

Overall, the **most common causes** of ear discharge are

- Acute (sudden and severe) middle ear infection ([otitis media](#)) with perforation ([puncture](#)) of the eardrum
- [Chronic otitis media](#) (with perforation of the eardrum, cholesteatoma, or both)
- External ear infection ([otitis externa](#))

In some people with otitis media (usually children), the eardrum ruptures, releasing the infected material collected behind the eardrum. The hole in the eardrum almost always heals, but sometimes a small perforation remains. A perforation may also result from injury or surgery to the eardrum. When a perforation is present, people are at risk of chronic middle ear infections, which can cause ear discharge.

Serious, but rare, causes of ear discharge include

- [Cancer of the ear canal](#)

- [Fracture of the base of the skull](#) with leakage of cerebrospinal fluid (CSF)
- [Necrotizing otitis externa](#) (previously called malignant otitis externa)
- Cholesteatoma (a noncancerous growth of skin material in the middle ear)

The ear canal passes through the base of the skull. If a skull fracture (from a severe head injury) involves that part of the skull, blood and/or cerebrospinal fluid may leak from the ear. If no head injury has occurred but CSF leaks from the ear, doctors may look for [idiopathic intracranial hypertension](#), which can cause a CSF leak.

Necrotizing otitis externa is a particularly severe form of external ear infection that typically occurs only in people with [diabetes](#), those who have a compromised immune system (due to, for example, [HIV infection](#) or [chemotherapy for cancer](#)), or the very old.

Some people with chronic otitis media develop a noncancerous (benign) growth of skin cells in the middle ear ([cholesteatoma](#)) that can cause discharge. Although a cholesteatoma is noncancerous, it can cause significant damage to the ear and nearby structures. In severe cases, a cholesteatoma may lead to deafness, balance loss, facial weakness or paralysis, and complications with the brain such as an abscess and other infections.

Evaluation of Ear Discharge

The following information can help people with ear discharge decide when a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with ear discharge, certain symptoms and characteristics are cause for concern:

- Recent major head injury
- Any neurologic symptoms (such as vertigo or difficulty seeing, speaking, swallowing, and/or talking)
- Hearing loss in the affected ear
- Fever
- Redness and/or swelling of the ear or area around the ear
- Diabetes or a compromised immune system

When to see a doctor

People with warning signs should see a doctor right away. People without warning signs should see a doctor as soon as possible and avoid getting water in the ear until it can be evaluated.

What the doctor does

In people with ear discharge, doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the ear discharge and the tests that may need to be done (see table

Some Causes and Features of Ear Discharge

During the medical history, doctors ask about the following:

- Activities that can affect the ear canal or eardrum (for example, swimming; insertion of objects, including cotton swabs; and use of ear drops)
- Whether people have had repeated ear infections
- Any severe head injury

During the physical examination, doctors focus on examining the ears, nose, throat, and neurologic system. By examining the ear canal with a light, doctors can usually diagnose perforated eardrum, otitis externa, foreign object, and other common causes of ear discharge. Other findings suggest the diagnosis.



Some Causes and Features of Ear Discharge

Cause	Common Features*	Diagnostic Approach
Acute discharge (lasting less than 6 weeks)		
<u>Acute otitis media with perforated eardrum</u>	Severe ear pain significantly relieved when a thick, whitish discharge starts	Sometimes doctor's examination alone Sometimes audiogram Sometimes high-resolution CT scan
<u>Chronic otitis media</u> with perforated eardrum	History of eardrum perforation and/or cholesteatoma (a noncancerous growth of skin cells in the middle ear), and previous discharge Eardrum appears abnormal during doctor's examination	Sometimes doctor's examination alone Sometimes high-resolution temporal bone CT Sometimes MRI
Cerebrospinal fluid leak caused by severe head injury or recent neurosurgery	Obvious recent head injury or neurosurgery Fluid ranges from crystal clear to blood	Imaging studies such as head CT scan including skull base or MRI with gadolinium
<i>Infectious:</i> Often after swimming or injury; severe pain, worse with pulling on ear		
<u>Otitis externa</u> (infectious or allergic)	<i>Allergic:</i> Often after use of ear drops; more itching and redness, and less pain than with infectious cause Typically a rash on the earlobe, where	Doctor's examination alone



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